

Subnetting Basic

1. Given the network 192.168.10.0/25, calculate the subnet mask, network address, broadcast address, and the range of usable IP addresses.

->For **192.168.10.0/25**:

1. Subnet Mask

/25 means 25 network bits and 7 host bits.

Subnet mask = 255.255.255.128

2. Network Address

The network starts at:

192.168.10.0

3. Broadcast Address

A /25 subnet has **128 addresses** (0–127).

So the last address is:

192.168.10.127

4. Usable IP Address Range

The network address and broadcast address cannot normally be assigned to hosts.

Therefore:

192.168.10.1 – 192.168.10.126

That's **126 usable IP addresses**.

Final Answer

Item	Result
Network	192.168.10.0/25
Subnet Mask	255.255.255.128

Item	Result
Network Address	192.168.10.0
Usable IP Range	192.168.10.1 – 192.168.10.126
Broadcast Address	192.168.10.127
Usable Hosts	126

genui{"learning_viz":{"type_id":"IPV4_SUBNETTING_CIDR","initial_values":{"address_octet_1":192,"address_octet_2":168,"address_octet_3":10,"address_octet_4":0,"prefix_length":25}}}

2. Your company wants to split the 10.0.0.0/24 network into four equal subnets for different departments (HR, Sales, Tech, Admin). List the network address, broadcast address, and usable IP range for each subnet.

->To divide **10.0.0.0/24** into **4 equal subnets**, we borrow **2 host bits**:

- $/24 + 2 = /26$
- New subnet mask: **255.255.255.192**
- Each subnet has **64 total addresses**
- **62 usable host addresses**

Department	Network Address	Usable IP Range	Broadcast Address
HR	10.0.0.0/26	10.0.0.1 – 10.0.0.62	10.0.0.63
Sales	10.0.0.64/26	10.0.0.65 – 10.0.0.126	10.0.0.127
Tech	10.0.0.128/26	10.0.0.129 – 10.0.0.190	10.0.0.191
Admin	10.0.0.192/26	10.0.0.193 – 10.0.0.254	10.0.0.255

Final subnet structure

10.0.0.0/24 → four /26 subnets

1. **HR:** 10.0.0.0 – 10.0.0.63
2. **Sales:** 10.0.0.64 – 10.0.0.127

3. **Tech:** 10.0.0.128 – 10.0.0.191

4. **Admin:** 10.0.0.192 – 10.0.0.255

Each department gets **62 usable IP addresses**.

3. A startup needs 60 usable IP addresses for its team. What is the smallest CIDR subnet you can assign, and what are the network address, broadcast address, and usable IP range for this subnet?

->To support **60 usable IP addresses**, we need to find the smallest subnet that provides at least 60 host addresses.

Step 1: Calculate host bits

Formula:

Usable addresses = $2^{\text{host bits}} - 2$

For 6 host bits:

$2^6 - 2 = 64 - 2 = 62$ usable addresses

So we need **6 host bits**.

CIDR = $32 - 6 = /26$

Subnet mask: **255.255.255.192**

Step 2: Choose a subnet

Assuming we use the common private range **192.168.1.0/24**, the first /26 subnet is:

- **Network address:** 192.168.1.0
- **Broadcast address:** 192.168.1.63
- **Usable IP range:** 192.168.1.1 – 192.168.1.62
- **Usable addresses:** 62

Final Answer

Item	Value
Smallest CIDR	/26

Item	Value
Subnet Mask	255.255.255.192
Network Address	192.168.1.0
Usable IP Range	192.168.1.1 – 192.168.1.62
Broadcast Address	192.168.1.63
Usable IPs	62

Therefore, a /26 subnet is the smallest subnet that can provide the required 60 usable IP addresses.

4. For the public WiFi at a Zomato restaurant, you are given the IP block 172.16.5.0/26.
Calculate the subnet mask, network address, broadcast address, and the first and last usable IPs.

Hint: /26 gives you 64 addresses per subnet.

->For the IP block **172.16.5.0/26**:

Step 1: Subnet Mask

/26 means 26 network bits and 6 host bits.

Subnet mask = 255.255.255.192

Step 2: Network Address

The given block starts at:

Network address = 172.16.5.0

Step 3: Broadcast Address

A /26 subnet contains **64 addresses**:

0–63

Therefore:

Broadcast address = 172.16.5.63

Step 4: Usable IP Range

The network and broadcast addresses cannot normally be assigned to hosts.

- **First usable IP:** 172.16.5.1
- **Last usable IP:** 172.16.5.62

Final Answer

Item	Answer
IP Block	172.16.5.0/26
Subnet Mask	255.255.255.192
Network Address	172.16.5.0
First Usable IP	172.16.5.1
Last Usable IP	172.16.5.62
Broadcast Address	172.16.5.63
Usable IPs	62

Note: 172.16.5.0 is actually part of the **private IPv4 range (172.16.0.0–172.31.255.255)**, so it would be appropriate for the internal network behind a public Wi-Fi router, not as a public Internet-facing IP block.

5. You receive the following scenario: Flipkart's warehouse needs three subnets supporting at least 30 devices each. Using CIDR notation, assign suitable subnets from the 192.168.1.0/24 range, and specify the network address, broadcast address, and usable IP range for each subnet.

->For **at least 30 devices per subnet**, we need a **/27** subnet:

- /27 = 32 total addresses
- Usable addresses = **32 – 2 = 30**
- Subnet mask = **255.255.255.224**

From **192.168.1.0/24**, we can create three **/27** subnets:

Warehouse Subnet	CIDR	Network Address	Usable IP Range	Broadcast Address
Subnet 1	192.168.1.0/27	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32/27	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64/27	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

Answer

Each subnet provides **30 usable IP addresses**, exactly meeting the requirement for 30 devices.

The remaining range **192.168.1.96–192.168.1.255** remains available for future subnets or expansion.